SMARTX: Porewater Sulfide

Sept 2025 Samples

2026-02-23

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed

sample_month = "September"
sample_year = "2025"
Run_Date = as.Date("09/19/2025", format = "%m/%d/%Y")
Run_by = "Melanie Giessner"#Instrument user
Script_run_by ="Brenden Blakley"#Code user

Run_notes=""#any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7", "Plate8", "Plate9")

Raw_plates <- c("Raw Data/20250919_SMARTX_H2S_Plate1.csv",
                "Raw Data/20250919_SMARTX_H2S_Plate2.csv",
                "Raw Data/20250919_SMARTX_H2S_Plate3.csv",
                "Raw Data/20250919_SMARTX_H2S_Plate4.csv",
                "Raw Data/20250919_SMARTX_H2S_Plate5.csv",
                "Raw Data/20250919_SMARTX_H2S_Plate6.csv",
                "Raw Data/20250922_SMARTX_H2S_Plate7.csv",
                "Raw Data/20250922_SMARTX_H2S_Plate8.csv",
                "Raw Data/20250922_SMARTX_H2S_Plate9.csv")

#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
    Plate = c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7","Plate8","Plate9"),
    Run_Date = c("09/19/2025","09/19/2025","09/19/2025","09/19/2025","09/19/2025","09/19/2025","09/22/2025
  )

  Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, format = "%m/%d/%Y"))


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/SMARTX_H2S_QAQC_2025.csv"
```

```r
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_SMARTX_Porewater_H2S_202509.csv"
rerun_path <- "Processed Data/GCReW_SMARTX_Porewater_H2S_202509_reruns.csv"
```

##Read in metadata and create similar sample IDs for matching to samples

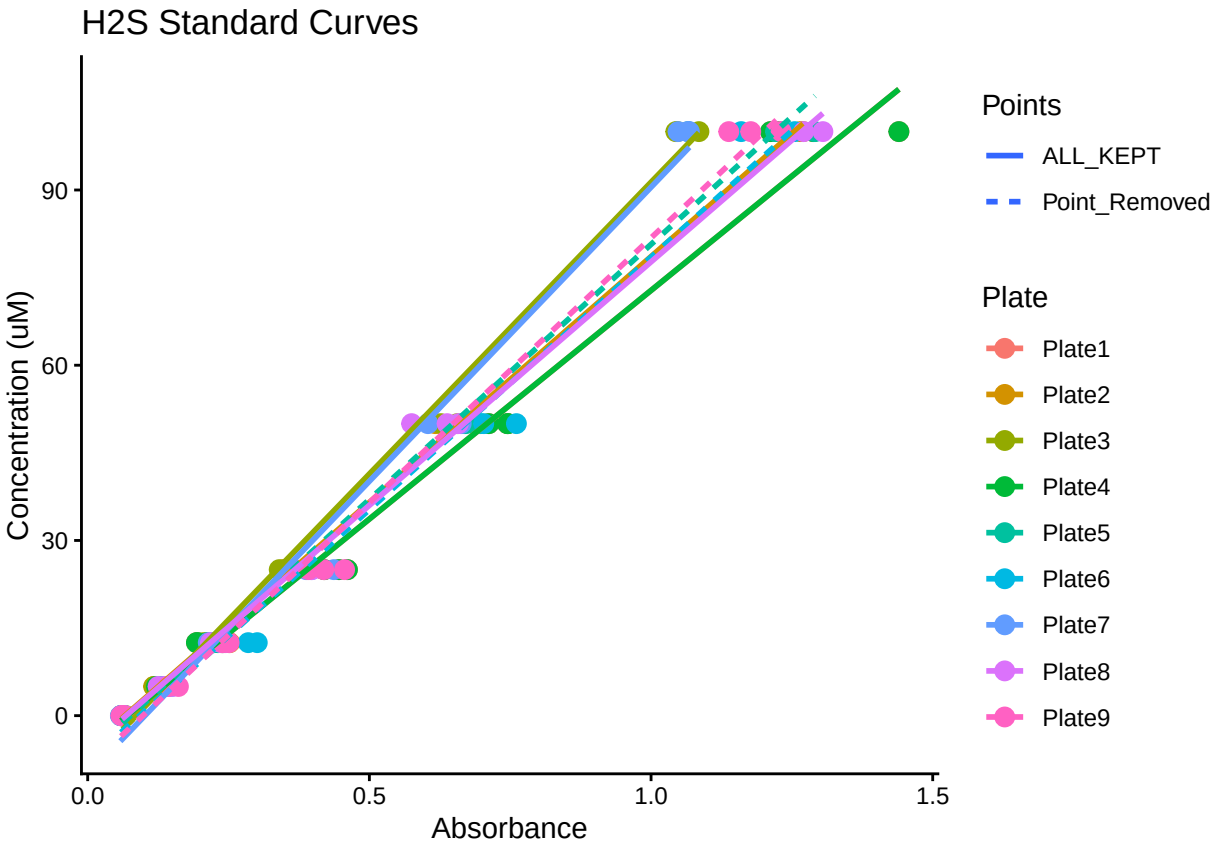
##Read in data

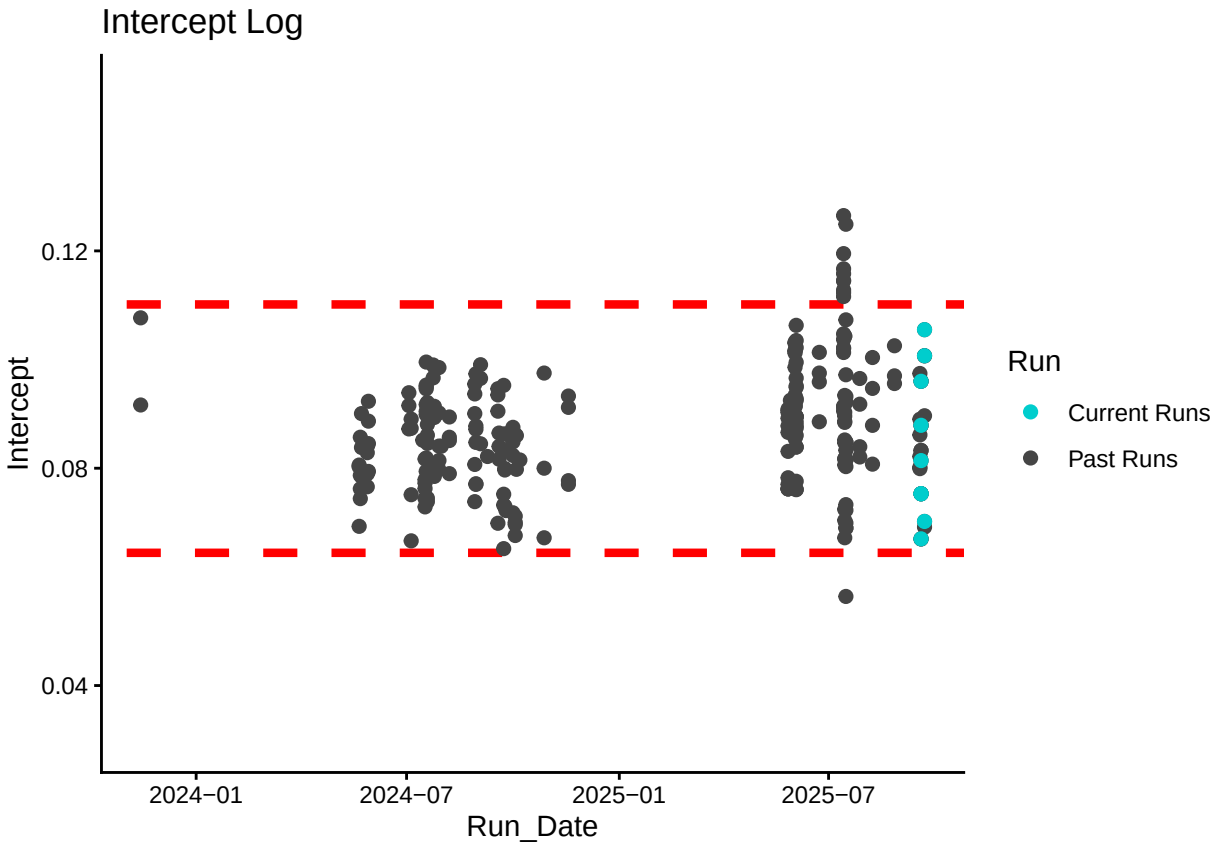
##Remove High CV Stds

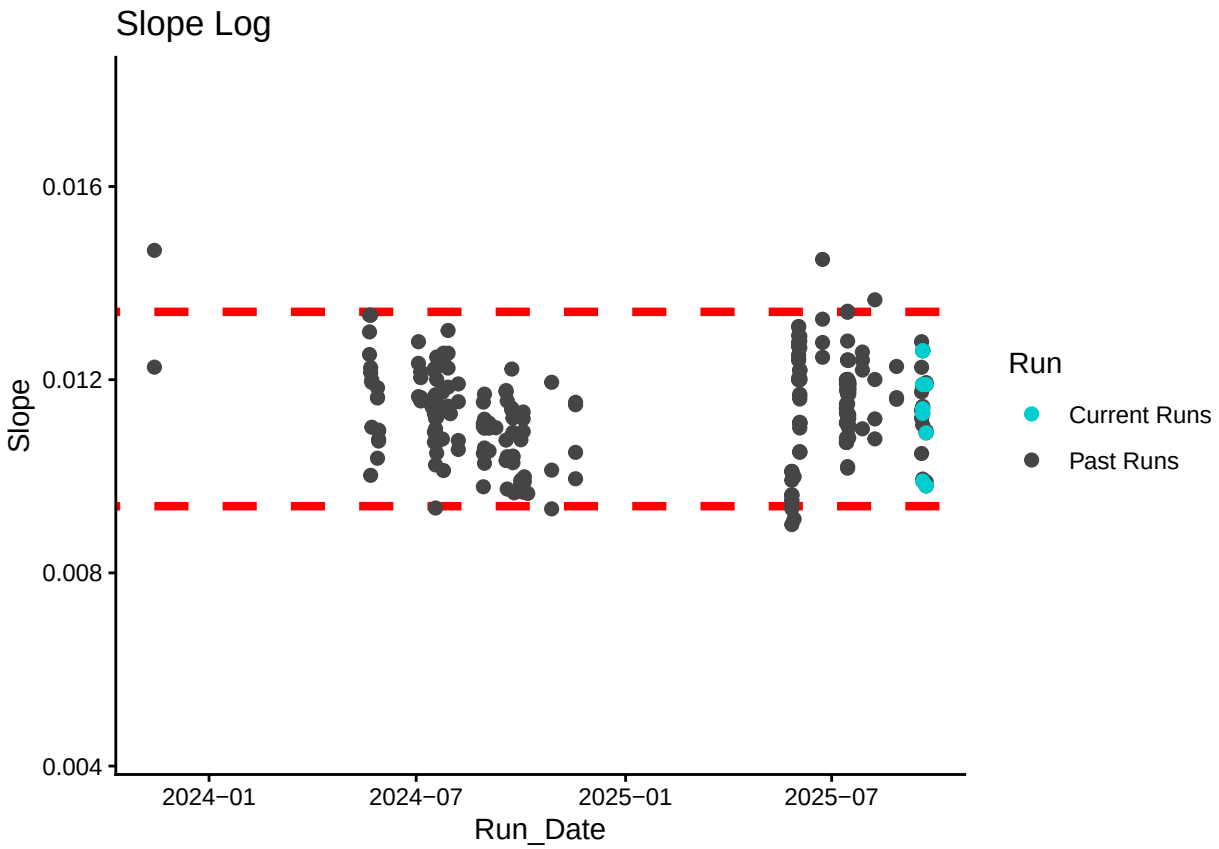
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

Plot Standard Curves





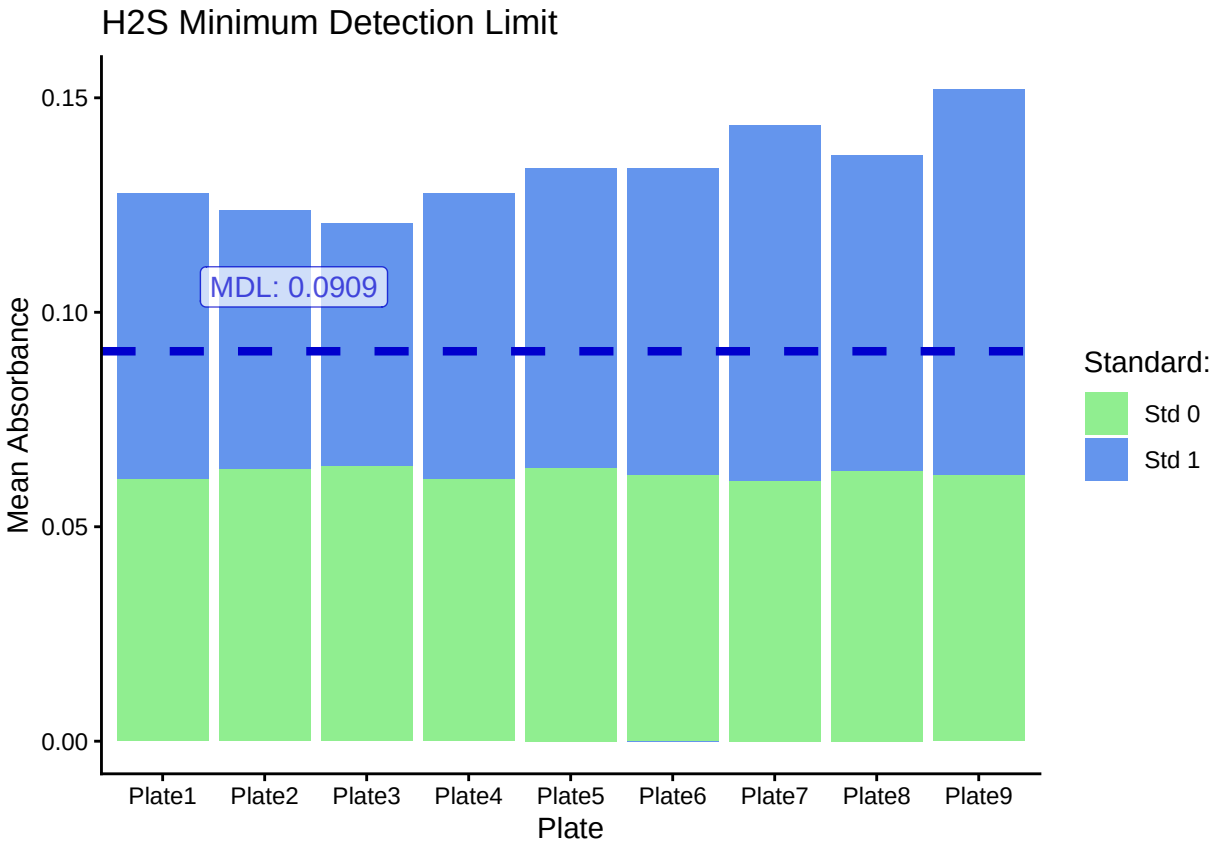


Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Good	CV_Good
Plate4	R2_Good	CV_Good
Plate5	R2_Good	CV_Good
Plate6	R2_Good	CV_Good
Plate7	R2_Good	CV_Good
Plate8	R2_Good	CV_Good
Plate9	R2_Good	CV_Good

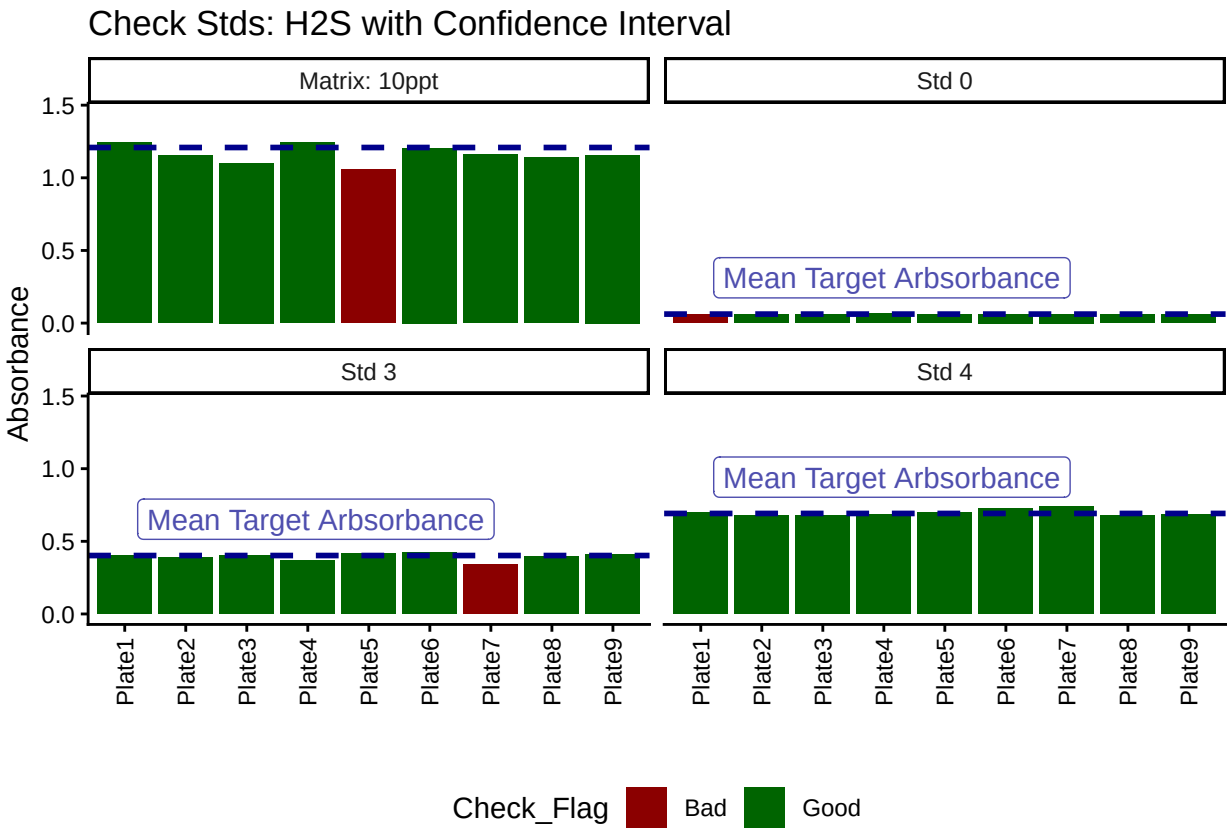
Plate	Curve_RMVD
Plate1	KEPT
Plate2	KEPT
Plate3	KEPT
Plate4	KEPT
Plate5	KEPT
Plate6	KEPT
Plate7	KEPT
Plate8	KEPT
Plate9	KEPT

Method Detection Limit



Compare Check Standards to Standards

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_label_repel()`).
```



Display Any Check Flags

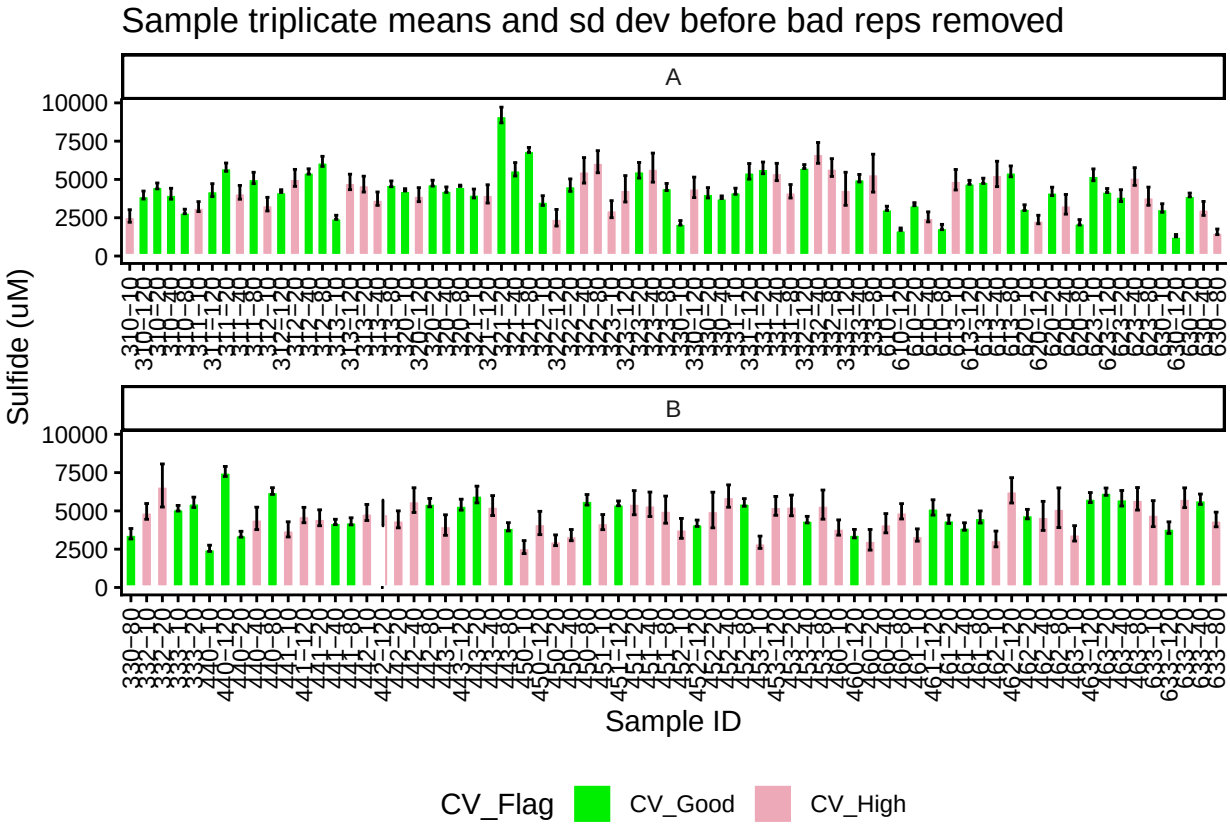
Plate	CHKs_Flag
Plate1	Std0Chk_Bad
Plate5	Std5Chk_Bad
Plate7	Std3Chk_Bad

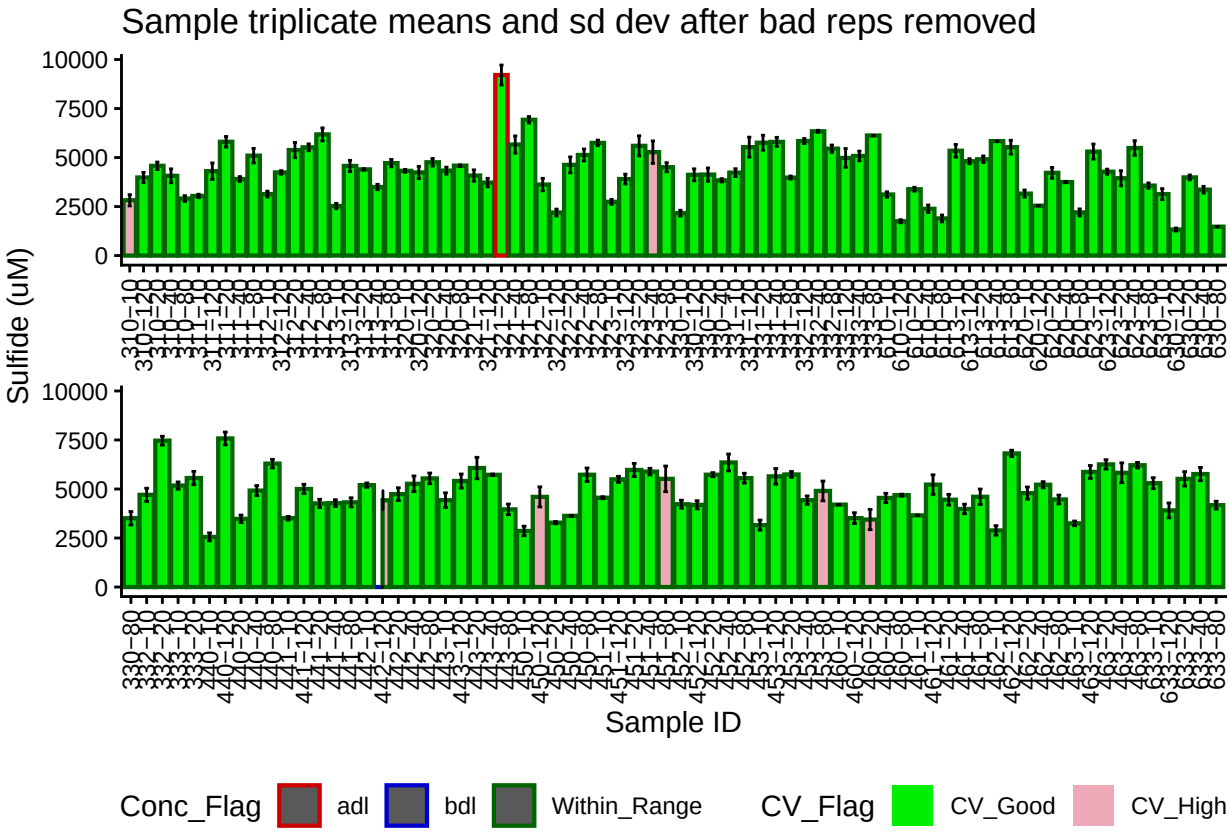
Plate	CHK_CV_Flags

##Calculate Sulfide Concentrations & Add Concentration Flags

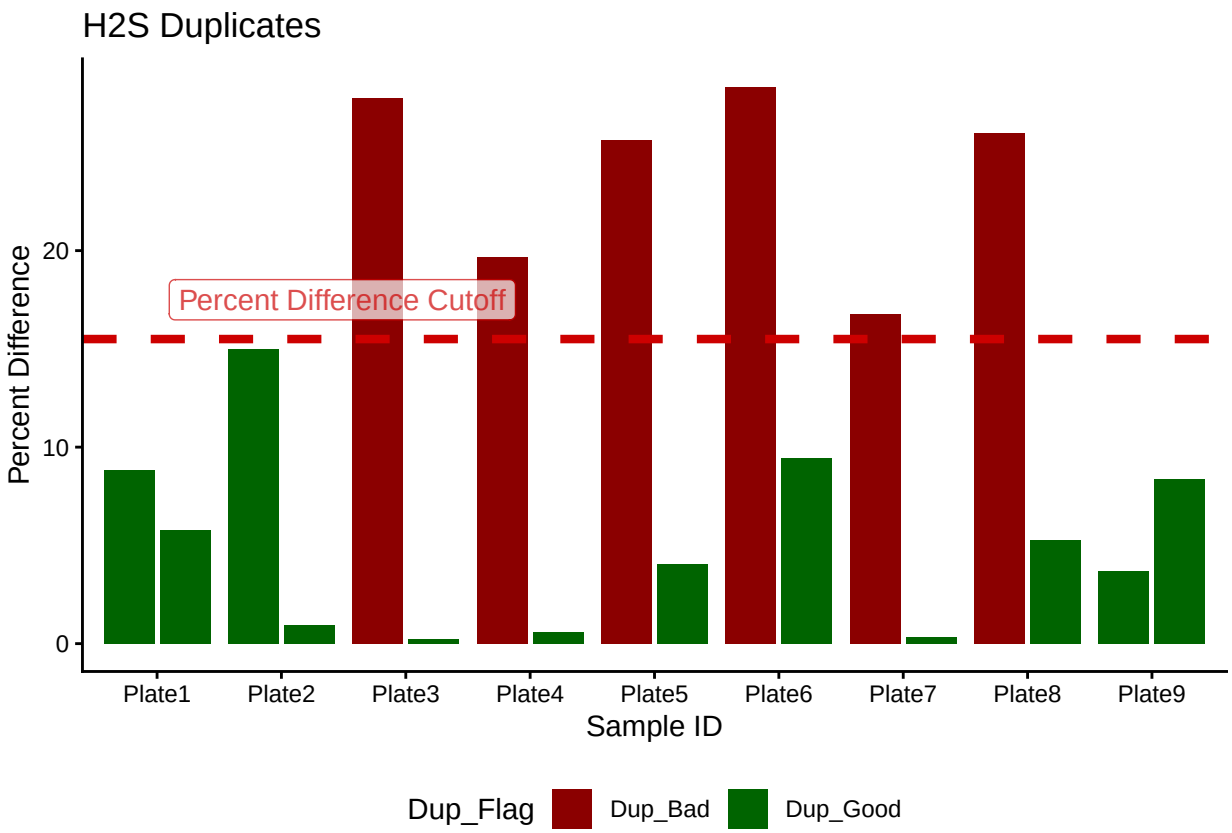
##Remove High CV Samples and Average

Plot Samples





Duplicate QAQC

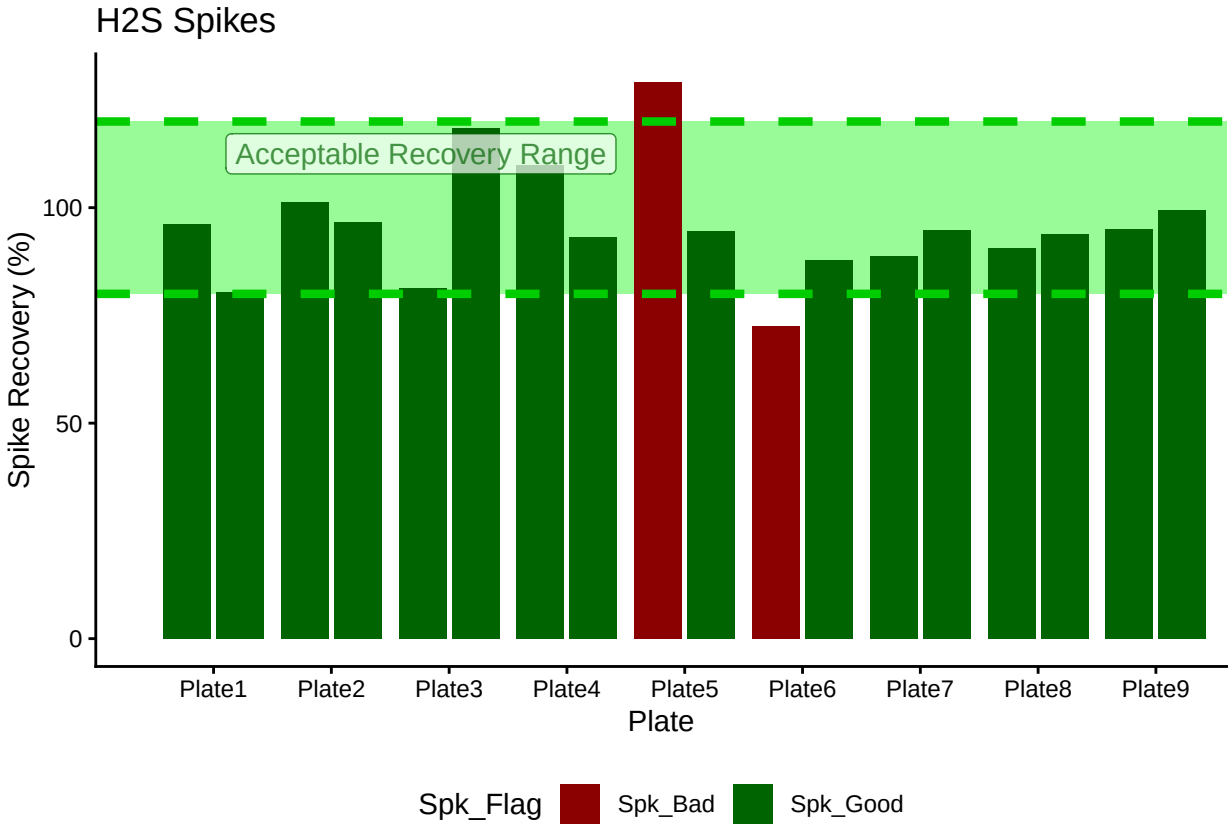


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Bad
Plate4	Dup_Bad
Plate5	Dup_Bad
Plate6	Dup_Bad
Plate7	Dup_Bad
Plate8	Dup_Bad
Plate9	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Good
Plate2	Spk_Good
Plate3	Spk_Good
Plate4	Spk_Good
Plate5	Spk_Bad
Plate6	Spk_Bad
Plate7	Spk_Good
Plate8	Spk_Good
Plate9	Spk_Good

```
## [1] "Spikes Pass"
```

Merge samples with Metadata & check all samples are present

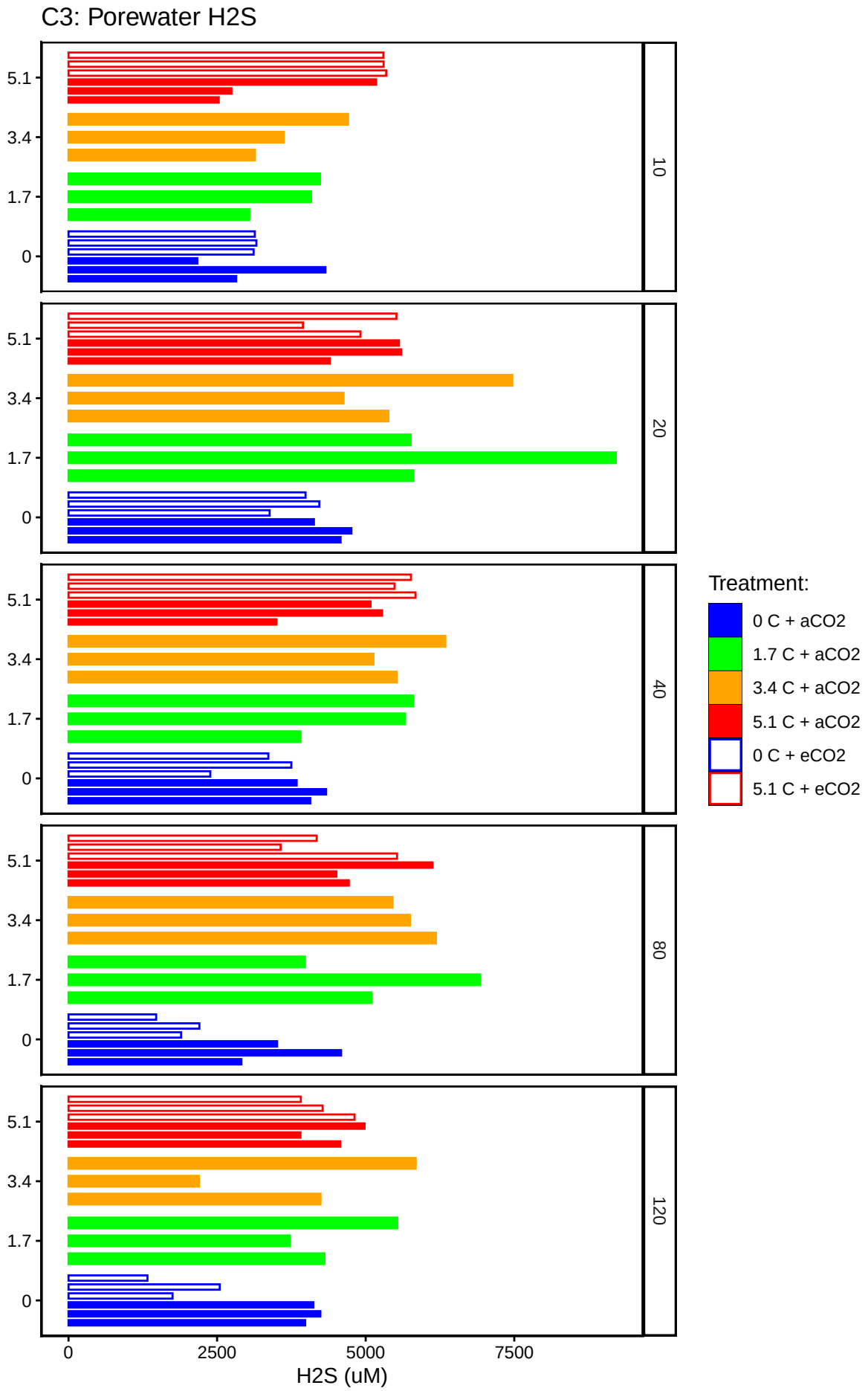
```
## All sample IDs are present in data
```

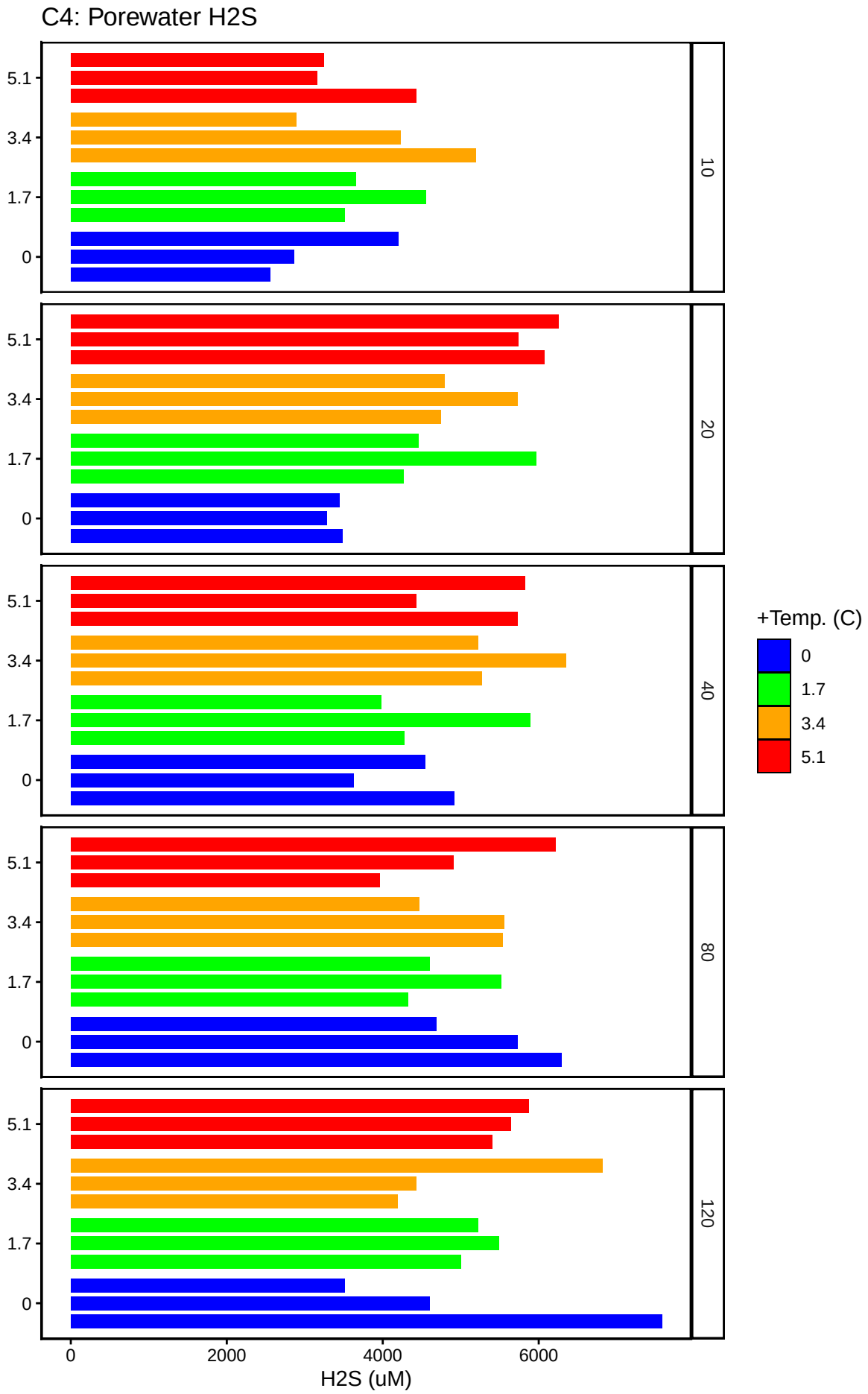
##Add Flags

##Organize Data

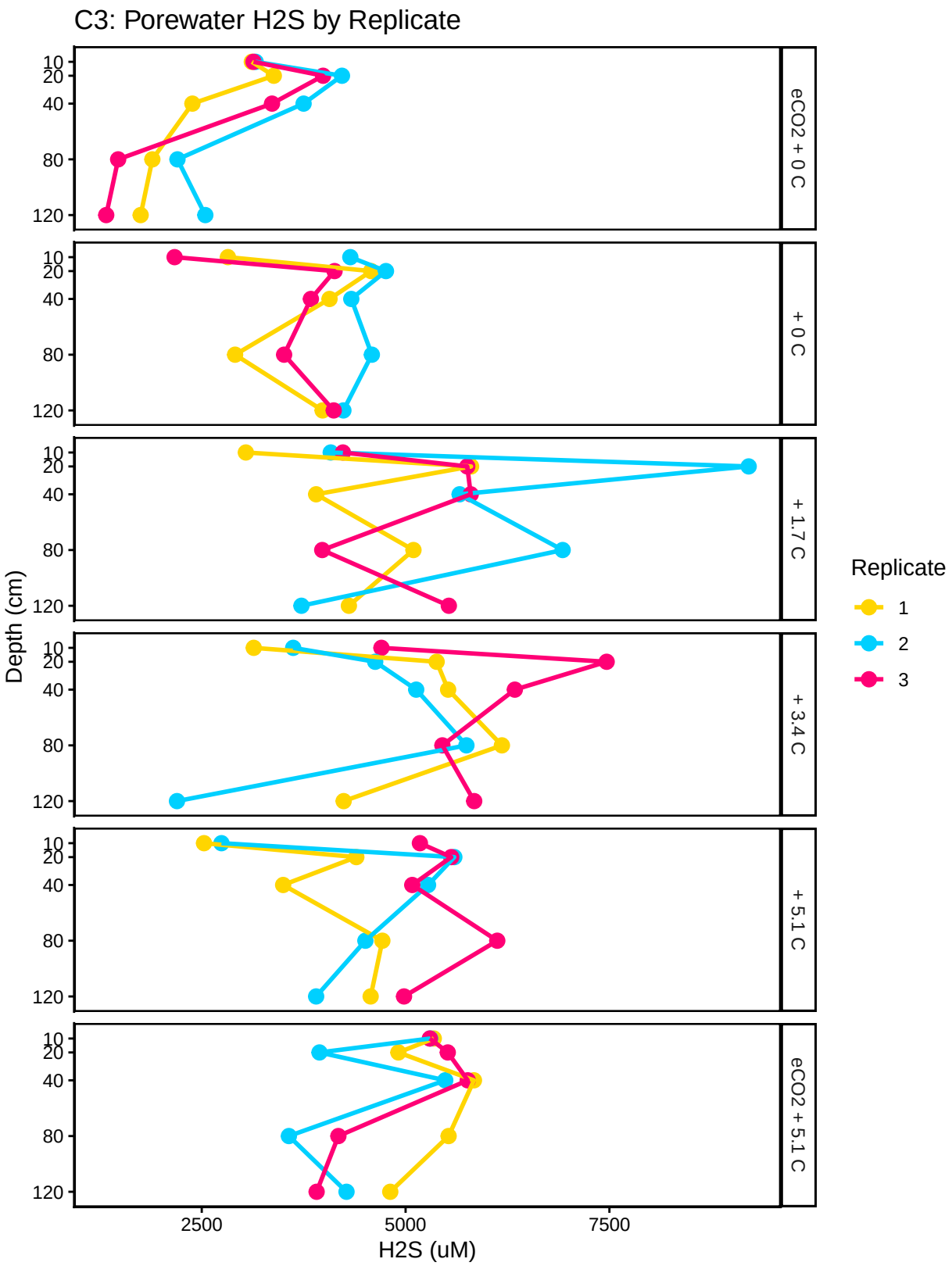
Visualize Data

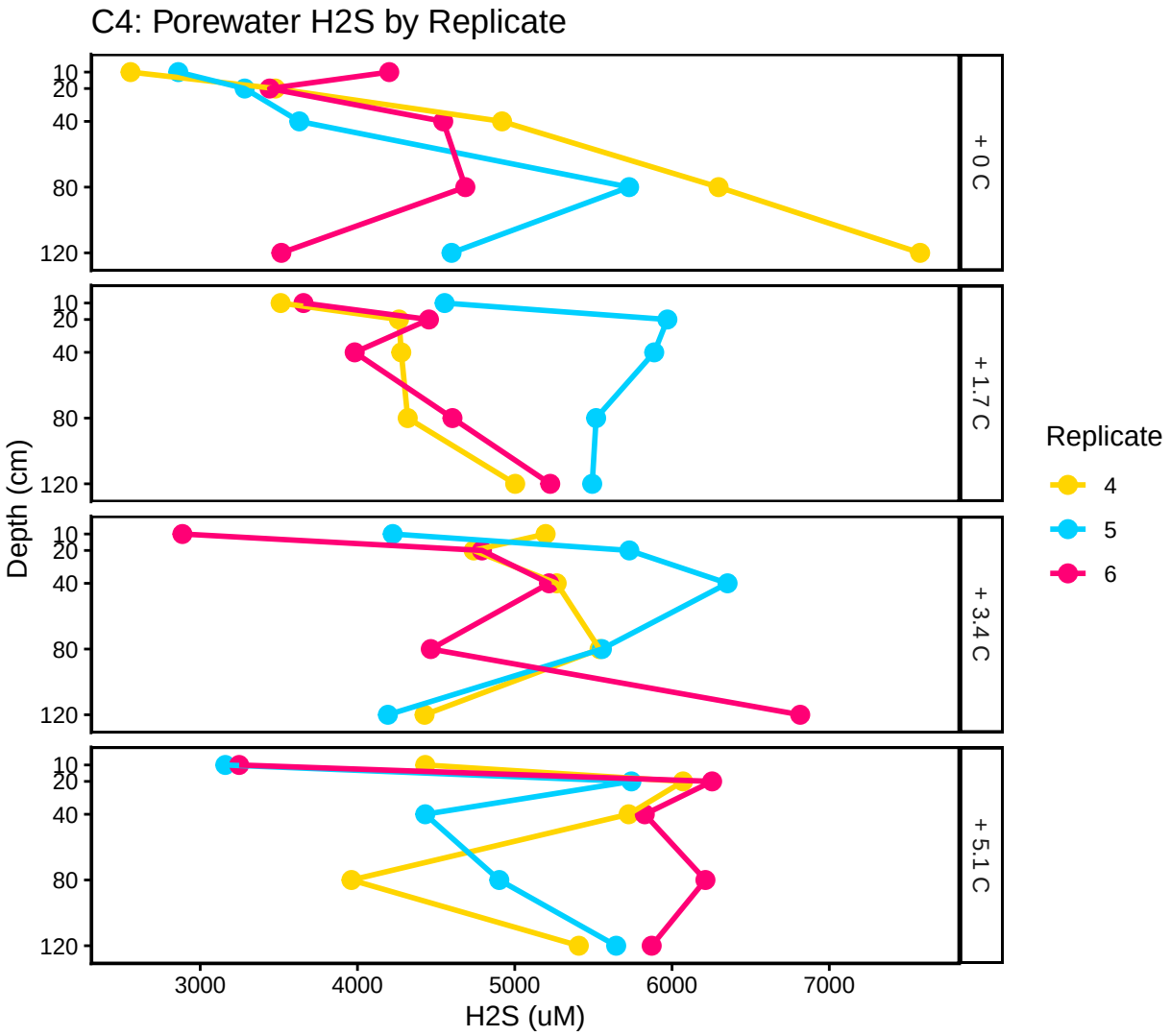
```
## Visualize Data
```


C3: Porewater H2S
5.1
3.4
1.7
0
10
20
40
80
120
0
2500
5000
7500
H2S (uM)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2


C4: Porewater H2S
5.1
3.4
1.7
0
10
20
40
80
120
0
2000
4000
6000
H2S (uM)
+Temp. (C)
0
1.7
3.4
5.1

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```





##Export Data

END